Introducing Quail: An Environment for Agentic Qualitative Analysis

Presenters

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Abstract

While text retrieval shows an agent a few examples, Quail gives it a Python kernel and an analysis language, in order to make sense of the whole dataset. Researchers often have more text than anyone can read closely [1]. LLMs can help, and one common way is through retrieval-augmented generation (RAG) [2], which gives a language model the few entries that best match a query. Standard RAG can find what was said about a topic but struggles with questions that require context over the whole dataset, like how often something was said and how it changed over time.

Notebooks like Jupyter let researchers iteratively explore. Quail gives an agent the same loop, where each cell (a block of Python) from the agent runs in a kernel that stays alive between cells. After each one, the agent receives printed output and the last line's value, just what a notebook would show. Many systems give an agent a separate tool for each operation. Quail instead relies on an analysis language inside its Python kernel, where functions to search, count, and tag are highly composable. Agents often work better this way than by calling tools one at a time [3], an approach now commonly called "code mode" [4].

Inside the Python kernel, Quail's analysis language acts as the bridge between the kernel and the dataset. The components of the language build on one another, from a dataset's columns to the four core functions. The language also gives a researcher and an agent a concise, shared vocabulary; this allows a researcher to efficiently interrogate an agent's actions.

Sessions in Quail are organized into studies. Because a study is a folder of text files, git can share it the way any piece of code is shared. The file structure is designed so that agents working in parallel can push logs without conflict and pick up where another agent left off. Semantic search needs the entire search space to be embedded, which can take hours to compute. `quail warm` can do this work ahead of time, distributed on several machines.

Operating Quail requires only a shell, so it can run wherever an agent does, on a laptop or a cloud VM. In one tested setup, TV news transcripts from six U.S. presidential election years, a Carleton College research dataset, were prepared as a study. Mentioning @Cursor in a Slack thread started a new Cursor cloud agent, whose VM booted from a snapshot that already held Quail, an embedding model, and the election news transcripts, indexed and embedded. Asked how Obama is characterized, the agent ran nearly 20 cells and replied 27 minutes later. Quail 0.94 is an early release for Linux and macOS, open source under the Apache-2.0 license [5].

References

1. The DataSquad Experiment: Lessons for Preparing Data and Computer Scientists for Work. Paula Lackie, Elliot Pickens, and Dashiell Coyier. arXiv:2511.19688. 2025. https://arxiv.org/abs/2511.19688
2. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Patrick Lewis et al. Advances in Neural Information Processing Systems 33. 2020. https://arxiv.org/abs/2005.11401
3. Executable Code Actions Elicit Better LLM Agents. Xingyao Wang et al. Proceedings of the 41st International Conference on Machine Learning, PMLR 235. 2024. https://arxiv.org/abs/2402.01030
4. Code Mode: the better way to use MCP. Kenton Varda and Sunil Pai. Cloudflare Blog. September 26, 2025. https://blog.cloudflare.com/code-mode/
5. Quail. Dashiell Coyier. GitHub repository. 2026. https://github.com/dcoyier/Quail

Connection to Mission, Goals, & Interests of US-RSE Community

This poster connects to US-RSE's Resources and Advocacy goals. For Resources, Quail is an open-source reference, under the Apache-2.0 license, for research software engineers building AI-assisted tools whose results researchers need to be able to check. The design decisions it presents come up in other projects as well: giving an agent a Python kernel with a small analysis language instead of a separate tool for each operation, so that a researcher can see exactly what it did; keeping the source data read-only; saving each cell to a log before the agent sees its result; and storing a study as text files that git can share and review.

For Advocacy, Quail is an example of what a campus data and software support group can build in response to a local need. It came out of the DataSquad, a student-staffed group at Carleton College, after we saw campus offices collect open-ended survey answers they did not have time to analyze. It also addresses this year's theme, Advancing Science in the Age of AI: in qualitative research, researchers need to see exactly what an agent did, and question it, before they can trust its analysis.

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